

National University of Computer and Emerging Sciences, Lahore Campus



Course:	Object Oriented Programming	Course Code:	CS217
Program:	BS(Computer Science)	Semester:	Fall 2019
Duration:	60 Minutes	Total Marks:	35
Paper Date:	24-Sep-2019	Weight	15
Section:	All	Page(s):	4
Exam:	Sessional-I	Roll No:	

1. Attempt all questions in the space provided in this sheet. You can use **rough sheets** but don't need to attach it here as it will not be marked.
2. Questions during exam are not allowed. Take reasonable assumptions where needed

Question 1:

Suppose you are given with the following **class fun** and two functions **print** and **function**. Write the output of the following code segment: [15 Marks]

```
class fun {
private:
    int x;
public:
    fun(int x1 = 0) {
        x = x1;
        cout << "constructor of ";
        print();
    }

    int getX() { return x; }
    void setX(int x1) { x = x1; }
    void print() {
        cout << " (" << x << ")" << endl;
    }
    fun(const fun & obj) {
        x = obj.x;
        cout << "Copy constructor of ";
        print();
    }
    ~fun() {
        cout << "destructor of ";
        print();
    }
};

void print(const int * p, int n) {
    for (int i = 0; i < n; i++)
        cout << p[i] << " ";
    cout << endl;
}
```

```
void main()
{
    cout << "Output (if any): " << endl;
    fun a(6);
    int list[6] = { 0,10,20,30,40,50 };
    int length = 3;
    int * array = &length;
    int * p = list;
    fun b = function(array, p, a);
    cout << "content of array: ";
    print(array, a.getX());
    cout << "content of p: ";
    print(p, (length / 2));
    cout << "content of list: ";
    print(list, length);
    cout << "Output (if any): " << endl;
}

_getch();
}
```

```
fun function(int*& arr, int * v, fun obj)
{
    *arr *= 2;
    int * var = v;
    v = (v + (obj.getX() / 2));
}
```

```

    for (int i = 0, j = 0; j < obj.getX(); j++, var++)
        *var = *(var + i) / 2;

    int k = obj.getX();
    obj.setX(2 * obj.getX());
    var--;
    arr = new int[k];

    for (int i = 0; i < k; i++) {
        arr[i] = i * k;
    }

    return obj;
} //function ends here

```

Answer:

Output (if any):	
content of array:	
content of p:	
content of list:	
Output (if any):	

Question 02:

Lapindrome is defined as a string which when split in the middle, gives two halves having the same characters and same frequency of each character. If there are odd number of characters in the string, we ignore the middle character and check for lapindrome. For example **gaga** is a lapindrome, since the two halves **ga** and **ga** have the same characters with same frequency. Also, **abccab**, **rotor** and **xyzxy** are a few examples of lapindromes. Note that **abbaab** is NOT a lapindrome. The two halves contain the same characters but their frequencies do not match.

Your task is simple, declare a C++ Class **Lapindrome** that has a function **checkLapindrome** which display Yes if the string is Lapindrome otherwise, it should display No. The structure of the Class is given below. You are required to implement every function given in the **class Lapindrome**.

[3+3+2+10+2=20 Marks]

```
class Lapindrome
{
private:
    int size;
    char* cstr;

public:
    Lapindrome(int size=10); //initialize size and cstr
    void setStr(char* cstr); //Assign the value of cstr to member variable
    cstr
    char* getStr();          //returns the string cstr.
    void checkLapindrome();  //Display Yes if the cstr is Lapindrome otherwise No
    ~Lapindrome();           //Destructor
};
```

